

Part II Compressible Flow

Objective:

The purpose of this part is to provide the student with the fundamental knowledge necessary to understand compressible aerodynamics.

Flow Regimes

$$M = \text{Mach number} = \frac{\text{velocity magnitude}}{\text{speed of sound}} = \frac{|\vec{u}|}{a}$$

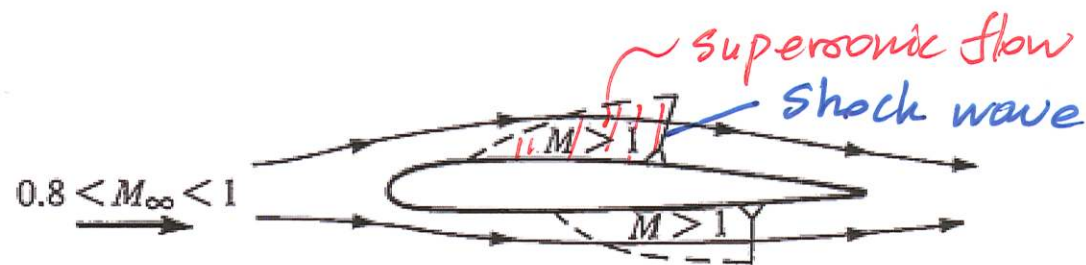
(Incompressible if $M < 0.3$)

$$a = \sqrt{\gamma R T}$$

(to be proven later)

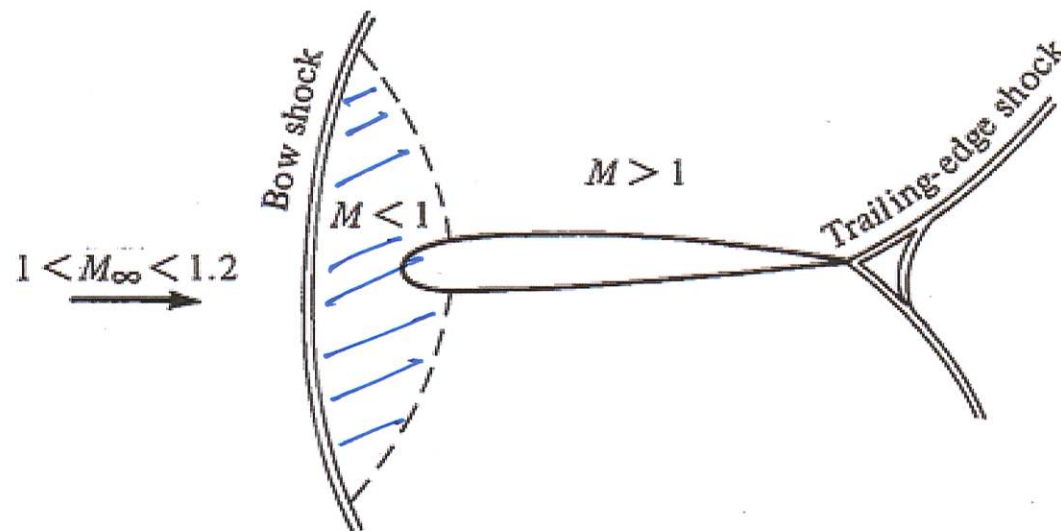


(a) Subsonic flow *every where*

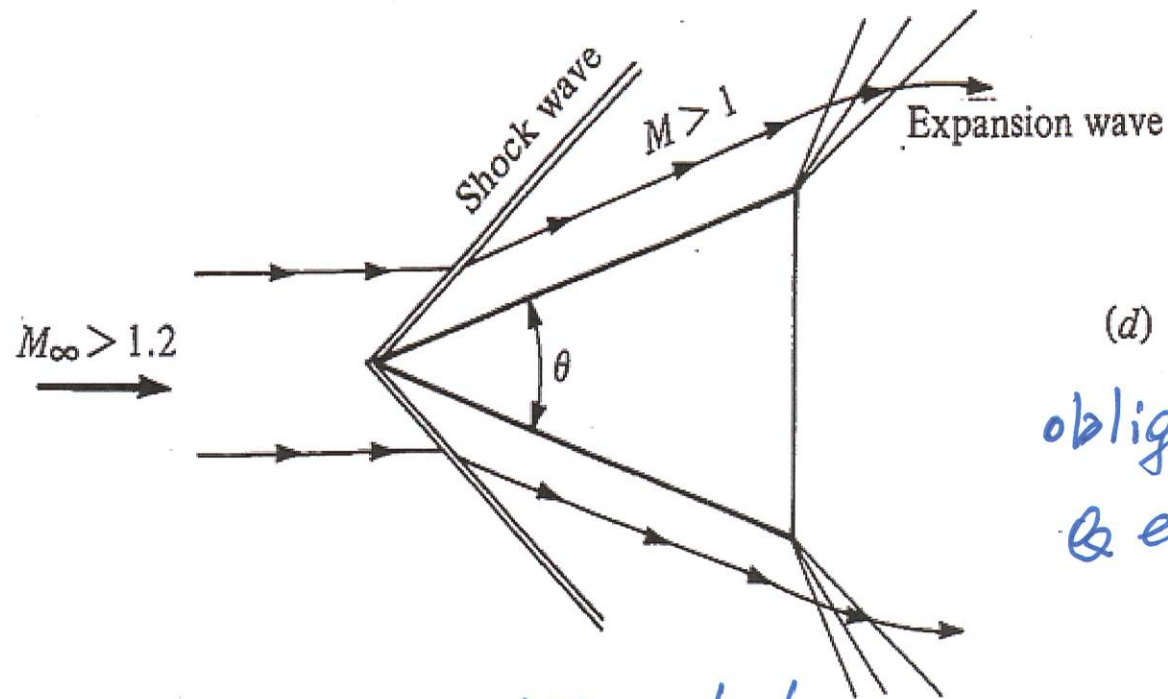


(b) Transonic flow
with $M_\infty < 1$

*regions of subsonic
& supersonic flow*

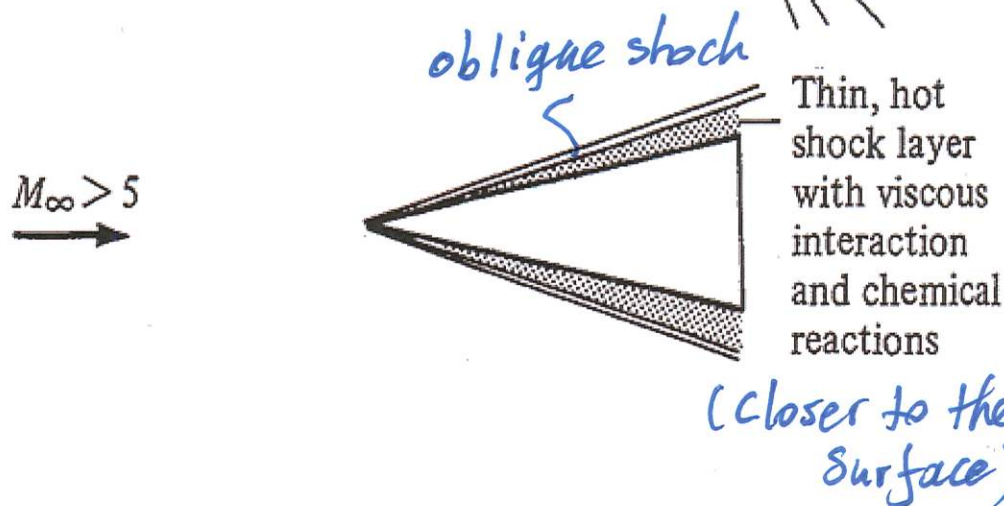


(c) Transonic flow
with $M_\infty > 1$



(d) Supersonic flow

oblique shocks
& expansion waves



(e) Hypersonic flow

- high temperature near body
possible chemical reactions dissociation
- discontinuous flow possible

$$M_n < M_\infty$$

→ reduce shock strength
→ reduce shock wave
drag

Why do commercial airliners have swept wings?



significant
drag
rise

What happens as you pass through the “sound barrier”



Why do supersonic airfoils have sharp edges?

- reduce shock wave drag
diamond shapes
at the exit

What types of flow can you have with a supersonic nozzle?



Dryden Flight Research Center EC92-1284 Photographed 1992
SR-71B take-off with "shock diamonds" in the exhaust. NASA photo



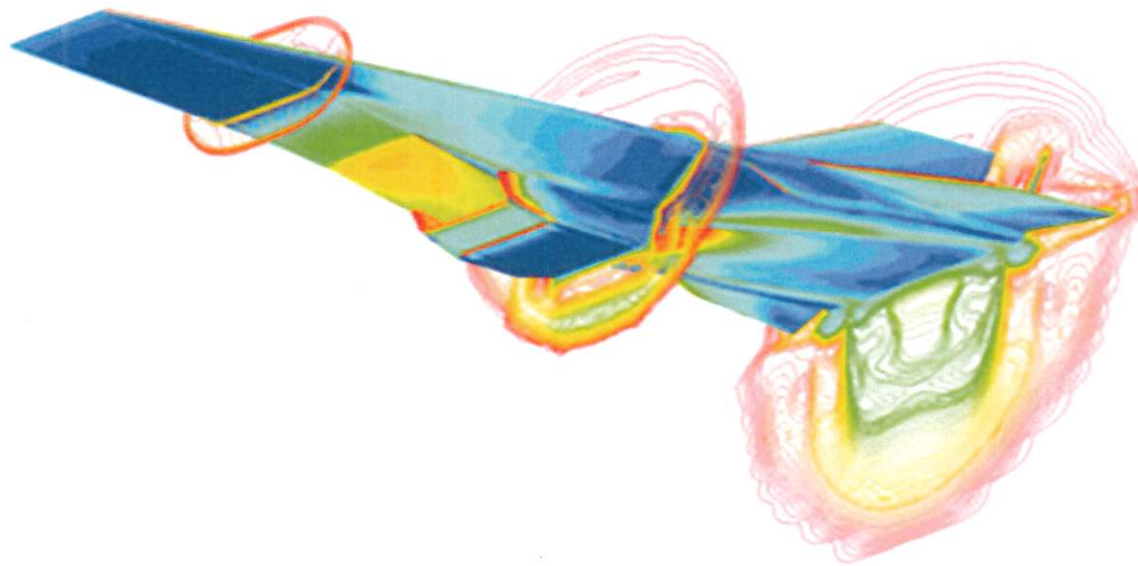
How do you analyze flow in a rocket nozzle?

$M < 1$ $M = 1$
converging/
diverging
nozzle to
achieve $M > 1$



http://www.concordesst.com/pictures/gboag_home.jpg

What happens at hypersonic speeds?



- shock is stronger
- near body
- heat becomes an issue
- wave drag

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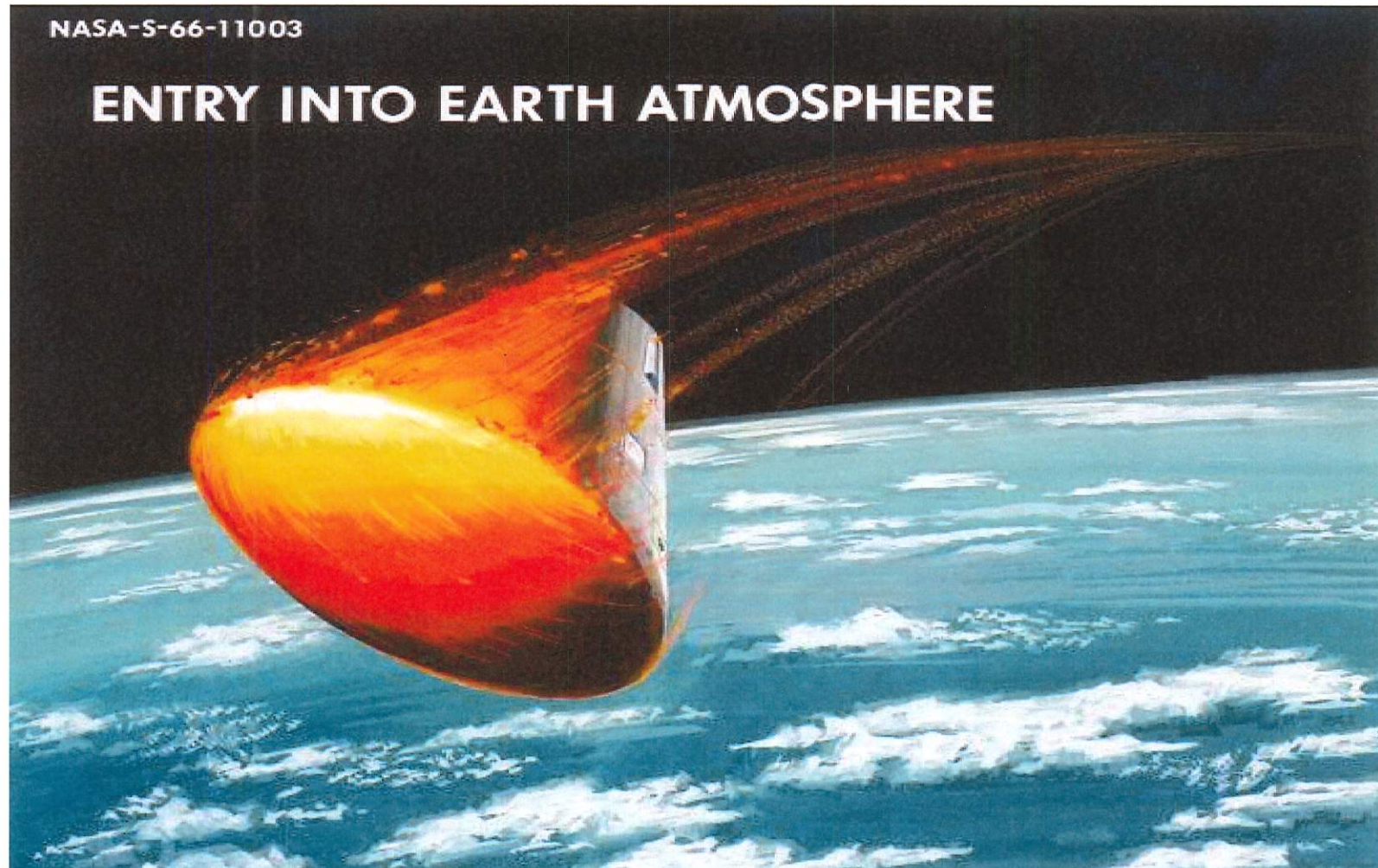


HYPER-X AT MACH 7: This computational fluid dynamic (CFD) image is of the Hyper-X vehicle at the Mach 7 test condition with the engine operating.



What happens during re-entry?

blunt body (heat) $\propto \frac{1}{\sqrt{Re_{e.e.}}}$



//
leading
edge
radius


small


big